
PageGauge: Factoring Affine KV Quantization Through Softmax Attention

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Abstract

1 KV-cache quantization is usually designed as a storage format and then repaired
2 inside the attention kernel by fused elementwise reconstruction. We instead co-
3 design the representation with the algebra of its consumer. PageGauge uses token-
4 independent channel centers and scalar page/head scales so that key centers cancel
5 as a softmax-invariant shift, value centers are restored once, and page scales
6 move outside the main code contractions. Small exact FP16 regions compose
7 with historical INT8 pages through online-softmax state merging. The transfor-
8 mation is exact in real arithmetic relative to the reconstructed mixed-precision
9 cache; quantization error remains distinct from floating-point execution error. On
10 Mistral-7B-v0.3 at batch four and a 20,480-token prefix, the final fixed policy re-
11 duces served KV storage by 36.86% and consumes 48 generated pages after they
12 age into INT8. Six fixed-policy WikiText-2 confirmation windows give minimum
13 PageGauge/FP16 logit cosine 0.99871 and top-1 agreement 0.99891. Eight fresh-
14 process Williams-order blocks measure 15.214 versus 17.232 ms per decode step,
15 a $1.1326\times$ end-to-end speedup (hierarchical 95% CI [1.1323, 1.1328]).

16 1 Introduction

17 Long-context autoregressive inference repeatedly streams a KV cache whose size grows with se-
18 quence length. Existing quantizers reduce storage, while fused kernels hide reconstruction within
19 attention [Liu et al., 2024, Hooper et al., 2024]. This can still move affine metadata through the
20 hottest contractions. Our starting axiom is broader: *quantization should be co-designed with the*
21 *algebra and invariances of the operation that consumes the tensor*. Softmax attention is unusually
22 favorable because it is invariant to a common key-logit shift, normalizes value weights to one, and
23 admits associative online-state merging [Dao et al., 2022, Dao, 2024].

24 We make exactly three contributions. (1) We give conditions under which the center and scale meta-
25 data of pagewise affine K/V quantization factor outside the main QK and PV code contractions. (2)
26 We realize this representation in a FlashInfer-compatible decoder path that directly consumes histor-
27 ical INT8 codes and merges them with exact FP16 regions without materializing a reconstructed
28 FP16 history. (3) We validate explicit-reconstruction fidelity, predictive fidelity with generated-page
29 recurrence, complete served-cache accounting, and repeated fresh-process end-to-end speed against
30 optimized FP16 FlashInfer [Ye et al., 2025]. We do not claim novelty for KV quantization, exact
31 recent windows, online softmax, or FlashInfer; the novelty is the representation/kernel co-design
32 that makes affine metadata factorizable.

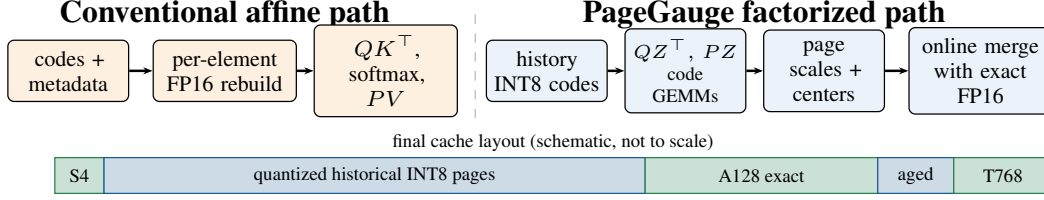


Figure 1: Conventional execution reconstructs historical affine K/V before attention. PageGauge contracts on code values, applies page/head scalar metadata at fragment granularity, handles common centers algebraically, and merges the result with centered exact regions. S4/A128/T768 denotes 4 exact prefix pages, 128 exact original-prefix suffix pages, and a 768-token exact recent tail.

2 Attention-Compatible KV Quantization

2.1 Affine representation and proposition

For one query and head, ordinary attention is $O = \text{softmax}(qK^\top/\sqrt{d})V$. For quantized page p , define

$$\hat{K}_p = \mathbf{1}c_K^\top + s_p^K Z_p^K, \quad \hat{V}_p = \mathbf{1}c_V^\top + s_p^V Z_p^V, \quad (1)$$

where Z_p is an INT8 code matrix, c_K, c_V are channel centers shared over the complete attention domain, and s_p^K, s_p^V are scalars over a page/head tile. Then

$$q\hat{K}_p^\top = (qc_K)\mathbf{1}^\top + s_p^K q(Z_p^K)^\top, \quad (2)$$

$$P\hat{V} = (P\mathbf{1})c_V^\top + \sum_p s_p^V P_p Z_p^V = c_V^\top + \sum_p s_p^V P_p Z_p^V. \quad (3)$$

The first term of Eq. (2) is common to every logit and therefore cancels under softmax. Each scalar scale is applied to a completed $qZ^\top$ or PZ fragment rather than to every stored element. Equation (3) restores the shared value center once because $P\mathbf{1} = 1$.

Proposition 1 (factorized mixed-cache attention). Suppose c_K, c_V are token independent and shared over the complete merged domain, and each s_p^K, s_p^V is scalar over its page/head contraction. If exact tokens use the same centered coordinates, PageGauge evaluates standard softmax attention over the reconstructed mixed-precision cache without elementwise affine reconstruction of historical quantized K/V. The equality is exact in real arithmetic relative to that reconstruction.

Proof sketch. Write exact tensors as $K'_E = K_E - \mathbf{1}c_K^\top$ and $V'_E = V_E - \mathbf{1}c_V^\top$. Concatenating $s_p^K Z_p^K$ with K'_E leaves one common $(qc_K)/\sqrt{d}$ shift, which softmax removes. Concatenating $s_p^V Z_p^V$ with V'_E yields one centered numerator; normalization restores c_V once. Associative merging below is algebraically identical to a single concatenated softmax. Full details and failure conditions are in Appendix A.

2.2 Mixed exact/quantized attention

Historical codes and exact centered regions are evaluated as separate stable online-softmax states (m_j, ℓ_j, o_j) . Their merge is

$$m = \max_j m_j, \quad \ell = \sum_j e^{m_j - m} \ell_j, \quad o = \sum_j e^{m_j - m} o_j, \quad O = o/\ell + c_V. \quad (4)$$

Thus a key-center offset is neither dropped per segment nor counted twice, and the value center is added only after global normalization. Figure 1 contrasts reconstruction-first execution with the factorized path.

2.3 Implementation and cost model

The production decoder stores page-major INT8 K/V codes and FP16 page/head scales, launches one historical-code wrapper and one exact-region wrapper, and uses FlashInfer’s stable merge.

Table 1: Final complete-decoder results. Fidelity is versus HF for FP16 and versus matched FP16 for PageGauge. CI is the hierarchical cache-neutral wall interval. “48 INT8” means generated pages aged out of the exact tail and were subsequently attended through the quantized path.

Backend	Served KV	Reduction / capacity	ms/step	Speedup (95% CI)	Min. cosine	Top-1	Recurrence
FlashInfer FP16	11.543 GB	–	17.2318	1.000×	0.999570	0.999132	FP16 reference
PageGauge S4/A128/T768	7.289 GB	36.86% / 1.584×	15.2141	1.1326× [1.1323,1.1328]	0.998707	0.998915	48 INT8 pages

Code values are converted to FP16 register arithmetic; we therefore avoid the unqualified phrase “dequantization-free.” Page finalization and device-dynamic graph planning remain inside the measured decode path.

With page size P , dimension d , and b -bit codes, FP16 K+V costs $4Pd$ bytes per page/head, whereas codes cost $2Pdb/8$ plus two scale values. At the 22,016-token endpoint, the final policy has 1,376 pages: 4 exact prefix pages, 128 exact original-prefill suffix pages, 48 exact recent-tail pages, and 1,196 quantized pages (86.92% of logical history). The implementation retains codes/scales even for exact pages, so the authoritative gross served cache is 7,288,520,704 bytes rather than an idealized compact count. A useful systems model is $S \approx [(1 - \alpha) + \alpha r + h]^{-1}$, where α is the FP16 decoder fraction affected by historical-KV attention, r its relative cost, and h merge/metadata/finalization overhead. It predicts a bounded full-model gain rather than the raw byte ratio; Section 3 measures that gain directly.

3 Evaluation

We evaluate Mistral-7B-v0.3 on one NVIDIA RTX 5090 (SM120), FP16 arithmetic, batch four, prefix 20,480, and 1,536 teacher-forced decode steps. Table 1 contains only final eligible results; protocol and source hashes are in the appendix.

Q1: Does factorization match explicit reconstructed attention? Across 27 old-only, exact-only, mixed-boundary, and ring cases spanning three split schedules, the explicit-reconstruction diagnostic passes every record; maximum absolute output error is 6.10×10^{-5} , minimum cosine is 0.999999821, and maximum LSE error is 9.87×10^{-5} . All 18 split-invariance checks pass. This isolates *factorization/execution error*. Differences from the original FP16 model instead reflect *quantization error*; GPU differences additionally reflect conversion, rounding, and accumulation order.

Q2: Does the fixed policy preserve predictive fidelity? Six disjoint WikiText-2 TEST windows (9,216 logits) give PageGauge/FlashInfer minimum cosine 0.998707, top-1 agreement 0.998915 (cluster-bootstrap lower 95% 0.998698), PPL-ratio upper 95% 1.000198, forward-KL p99 6.05×10^{-5} , and JS p99 1.51×10^{-5} . Each run verifies that 48 generated pages were finalized and later consumed as INT8. This is a fixed-policy confirmation, not untouched held-out selection: a preregistered S3/T768 policy failed its TEST minimum-cosine gate, and S4/A128/T768 was fixed after diagnosing the exposed failure. Rare local KV errors can be amplified by later blocks, motivating the small exact robustness regions without making them a novelty claim.

Q3: How much served KV storage is removed? Matched FP16 storage is 11,542,724,608 bytes; PageGauge uses 7,288,520,704 bytes including exact K/V, gross code tensors, scales, and centers. The saving is 4,254,203,904 bytes (36.856%), or $1.5837\times$ effective capacity. Exact regions alone use 1.508 GiB of FP16 K+V; no compact-storage credit is taken for their retained INT8 slots.

Q4: Does the reduction improve end-to-end speed? Eight fresh processes use ABBA then BAAB order over two disjoint fixture/seed clusters. Four adjacent backend pairs, three repeats per block, and 50,000 paired and hierarchical bootstrap draws yield 17.231814 ms/step for FlashInfer and 15.214065 for PageGauge. The primary cache-neutral wall speedup is $1.132624\times$; hierarchical 95% CI is [1.132340, 1.132797] and paired-block CI [1.132358, 1.132788]. AB and BA strata are $1.132496\times$ and $1.132752\times$. Model load, prefill, graph capture, scrub, and restoration are excluded symmetrically; the interval includes 32 decoder layers, page planning, embedding, final norm, LM head, and GPU argmax.

Policy realization and recurrence. The endpoint partition is explicit rather than inferred from a nominal tail length. Pages 0–3 are an exact 64-token prefix; pages 4–1151 are quantized original history; pages 1152–1279 are an exact 2,048-token original-prefix suffix; generated pages 1280–1327 have aged into INT8; and pages 1328–1375 form the 768-token exact recent tail. Prefix and suffix pages are immutable, whereas new pages close every 16 steps, enter the ring, and are quantized when displaced. The quality and performance artifacts verify disjoint/exhaustive page tables, immutable exact-region digests, finite positive scales, all 96 page closures, and attention to the oldest 48 generated pages through the INT8 wrapper. Recurrence is therefore an executed cache-state property, not a byte formula alone.

Validity controls and eligible evidence. Every timed backend first passes same-backend eager-versus-graph checks for logits, generated tokens, cache tensors, page-close digests, and planner state. Each performance block then starts a new process holding the model and only one backend cache; an earlier co-resident experiment near VRAM capacity is excluded because allocation placement created an artificial late-layer cliff. The final analyzer requires matched corpus, model, teacher trajectory, graph scope, source hashes, memory geometry, and execution counters before forming an adjacent pair. The explicit-reconstruction result in Q1 uses an experimental heterogeneous wrapper to test the algebra across boundary/ring cases; it is appropriately a factorization diagnostic, not a timing surrogate for the final segmented path.

Interpreting capacity and speed. The $1.5837\times$ capacity ratio is deliberately not reported as a latency prediction. Historical attention is only a fraction α of a full layer, the unaffected term $(1 - \alpha)$ remains, and PageGauge adds online merge, metadata, and page-finalization work. The measured $1.1326\times$ complete- decoder gain is consequently smaller than both the storage ratio and isolated resident-attention gains, as the cost model predicts qualitatively. Conversely, the narrow two-cluster interval establishes chronology-stable improvement for these matched fixtures, not a universal hardware- or workload-level effect.

4 Limitations and Conclusion

The algebra generalizes under its stated metadata assumptions, but measurements cover one Mistral-7B configuration, batch/context regime, and RTX 5090. We have not established cross-model or cross-GPU transfer. PageGauge converts codes to FP16 arithmetic, exact regions have a material byte cost, and the two-cluster performance CI does not express broad workload uncertainty. The benchmark is a complete decoder with sustained page planning, not multi-tenant production serving with networking, admission, or prefill latency. Most importantly, the final precision policy followed TEST-visible adaptation and needs independent future-corpus confirmation. Within that scope, PageGauge supports the central axiom: choosing quantization metadata to match a consumer’s invariances can turn reconstruction work into cheap fragment-level corrections and deliver a measured end-to-end gain.

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A Extended Derivation and Assumptions

This appendix makes the coordinate and merge assumptions explicit. Fix a request, transformer layer, and KV head. The channel centers c_K, c_V are shared across every token in the complete attention domain. Quantized page p is represented by

$$\widehat{K}_p = \mathbf{1}c_K^\top + s_p^K Z_p^K, \quad \widehat{V}_p = \mathbf{1}c_V^\top + s_p^V Z_p^V,$$

with scalar s_p^K, s_p^V over the page/head tile. Exact tokens are stored in centered form $K'_E = K_E - \mathbf{1}c_K^\top$ and $V'_E = V_E - \mathbf{1}c_V^\top$. Define the centered reconstructed cache

$$\widetilde{K} = [\{s_p^K Z_p^K\}_{p \in \mathcal{Q}}, K'_E], \quad \widetilde{V} = [\{s_p^V Z_p^V\}_{p \in \mathcal{Q}}, V'_E].$$

Then $\widehat{K} = \mathbf{1}c_K^\top + \widetilde{K}$ and $\widehat{V} = \mathbf{1}c_V^\top + \widetilde{V}$. For one query,

$$\text{softmax}\left((q\widehat{K}^\top)/\sqrt{d}\right) = \text{softmax}\left((q\widetilde{K}^\top)/\sqrt{d}\right)$$

because $(qc_K)/\sqrt{d}$ is common to every logit. If the resulting row vector is P , then

$$P\widehat{V} = P\mathbf{1}c_V^\top + P\widetilde{V} = c_V^\top + P\widetilde{V}.$$

The claim fails if centers differ between independently evaluated segments and their offsets are not carried into the global merge. It also requires scalar page/head scales; per-channel scales generally cannot move outside the dot product in the same way.

For multiple segments j , online softmax maintains a maximum m_j , an exponential denominator ℓ_j , and centered numerator o_j . The associative merge is

$$m = \max_j m_j, \quad \ell = \sum_j e^{m_j - m} \ell_j, \quad o = \sum_j e^{m_j - m} o_j, \quad O = o/\ell + c_V.$$

This is the same real-arithmetic result as a single centered softmax over the concatenated domain. FlashInfer’s split-K merge supplies the same stable-state structure; the final production path exposes two logical attention segments.

B Quantization Error Versus Execution Error

Two comparisons must not be conflated:

1. **Quantization error:** original FP16 (K, V) versus reconstructed $(\widehat{K}, \widehat{V})$.
2. **Factorization/execution error:** explicit FP16 attention over the reconstructed cache versus the factorized PageGauge execution.

The proposition eliminates the second in real arithmetic. GPU kernels can still differ through FP16 conversion, rounding, accumulation order, split-K partitioning, or the location of scalar multiplication.

The explicit-reconstruction diagnostic uses nine old-only, exact-only, boundary, and ring-wrap cases, each under automatic, fixed-2, and fixed-8 split settings. All 27 cases pass. Across candidate versus reconstructed FP16 FA2, maximum absolute output error is 6.1035×10^{-5} , minimum cosine is 0.999999821, maximum relative ℓ_2 is 4.2093×10^{-4} , and maximum LSE absolute error is 9.8705×10^{-5} . All 18 cross-split comparisons pass. This artifact exercises an experimental heterogeneous wrapper, so we use it as a development diagnostic of the algebra and implementation pattern, not as a speed result for the final segmented system. The final full-decoder blocks add same-backend eager-versus-graph bitwise logits/cache checks, while the separate quality cohort measures quantization plus recurrence end to end.

C Final Precision Policy and Byte Accounting

At the endpoint, each request represents 22,016 tokens in 1,376 pages of 16. The fixed policy is:

- exact prefix: pages 0–3, 64 tokens;
- quantized original history: pages 4–1151;
- exact fixed original-prefill suffix: pages 1152–1279, 2,048 tokens;
- generated pages aged into INT8: pages 1280–1327, 768 tokens;
- exact recent tail: pages 1328–1375, 768 tokens.

Thus 1,196/1,376 pages, or 86.9186% of the represented logical history, are quantized at the end-point. The exact logical fraction is 13.0814%. The exact regions are a robustness allocation, not a novelty claim.

For one page/head with $P = 16$, $d = 128$, and $b = 8$, FP16 K+V costs $4Pd = 8,192$ bytes. INT8 codes cost $2Pd = 4,096$ bytes, plus two FP16 scalar scales (4 bytes); centers are shared across pages. The implementation does not compact away code/scale storage for exact pages. The raw served-cache tensors, excluding one following-canary page, are:

Tensor class	Bytes	Note
Exact K	754,974,720	180 pages/request
Exact V	754,974,720	180 pages/request
K codes	2,885,681,152	gross 1,376 pages/request
V codes	2,885,681,152	gross 1,376 pages/request
K scales	2,818,048	served pages only
V scales	2,818,048	served pages only
Centers/output center	1,572,864	request/layer/head tensors
PageGauge served total	7,288,520,704	authoritative geometry
FlashInfer FP16 total	11,542,724,608	authoritative geometry

The source allocation includes a following canary and therefore prints 7,292,719,104 bytes before subtracting 4,198,400 canary bytes. PageGauge saves 4,254,203,904 bytes (3.962 GiB), a 36.8562% reduction and $1.58369\times$ effective capacity.

D Predictive-Quality Protocol and Adaptive Disclosure

The original source-frozen selection was S3/T768 with SHA-256 D1C8675B3B1F70437023DC8CD56665EA79602AAF6EDA2752717C0CEBC127798F. Its preregistered TEST shard failed the minimum-cosine gate. Diagnostics found a separate HF-conversion numerical mismatch and, after repairing it, a residual PageGauge quantization error amplified across layers. The final S4/A128/T768 policy was chosen after inspecting exposed TEST behavior. We therefore do not describe the final six-window result as untouched held-out selection.

The final fixed-policy confirmation uses six disjoint WikiText-2 raw TEST windows at offsets 0, 23,600, 47,200, 70,800, 94,400, and 118,000. Each window contributes 1,536 labeled logits, for 9,216 total. The reducer resamples six whole windows, not tokens, for 50,000 draws. All PageGauge/FlashInfer gates pass:

Metric	Estimate	Gate
Minimum cosine	0.9987073	≥ 0.995
Top-1 agreement	0.9989149	≥ 0.99
Top-1 lower 95%	0.9986979	≥ 0.98
PPL-ratio upper 95%	1.0001982	≤ 1.01
Maximum window PPL ratio	1.0002579	≤ 1.03
Mean forward-KL upper 95%	8.09×10^{-6}	$\leq 10^{-3}$
Forward-KL p99	6.05×10^{-5}	$\leq 10^{-2}$
Mean JS upper 95%	2.02×10^{-6}	$\leq 2.5 \times 10^{-4}$
JS p99	1.51×10^{-5}	$\leq 2.5 \times 10^{-3}$

FlashInfer/HF SDPA also passes its tighter gates. Each run finalizes 96 generated pages and verifies that the oldest 48 are present in the INT8 attention domain.

216 E Fresh-Process Performance Protocol

217 The final performance experiment runs eight independent Python/CUDA processes. The first fix-
 218 ture/seed uses ABBA order; the second disjoint fixture/seed uses BAAB. Adjacent backend blocks
 219 form four pairs, and each block contains one warmup and three repeats. Each process loads the
 220 model and only its selected backend cache; load, prefill, graph capture, cache restoration, scrub, and
 221 preconditioning are excluded. The measured interval contains all 1,536 continuous steps, 32 com-
 222 plete decoder-layer graph replays per step, page-table and FlashInfer planning updates, embedding,
 223 final norm, LM head, and GPU argmax.

224 The primary estimand is

$$\exp \left(\frac{1}{4} \sum_{i=1}^4 \log \frac{t_{\text{FI},i}}{t_{\text{PG},i}} \right).$$

225 The paired-block bootstrap resamples the four adjacent pairs. The hierarchical bootstrap first resam-
 226 ples the two fixture/seed clusters, then the two pairs within each selected cluster. Both use 50,000
 227 draws. Cache-neutral wall results are:

Quantity	Value
FlashInfer geometric mean	17.231814 ms/step
PageGauge geometric mean	15.214065 ms/step
Speedup	1.132624×
Paired-block 95% CI	[1.132358, 1.132788]
Hierarchical 95% CI	[1.132340, 1.132797]
AB order stratum	1.132496×
BA order stratum	1.132752×

228
 229 Cache-hot wall speedup is 1.132693× with hierarchical 95% CI [1.132086, 1.133083]. The nar-
 230 row intervals describe these two fixtures and their repeat chronology; they do not establish cross-
 231 workload uncertainty.

232 The raw workers report overall false because their status also incorporates a strict cross-HF timing-
 233 fixture diagnostic. This does not make a failed execution block eligible: the final analyzer separately
 234 requires every same-backend eager/graph replay, cache, page-table, canary, recurrence, source, and
 235 shared identity gate. All eight performance execution gates pass. Predictive fidelity is supplied by
 236 the independent six-window full-distribution aggregate.

237 F Development Diagnostics and Negative Evidence

238 Several results are deliberately excluded from headline evidence. A co-resident full-model exper-
 239 iment held both full caches near device capacity and showed an artificial late-layer slowdown;
 240 fresh allocation/relocation removed the cliff. A heterogeneous two-node attention wrapper matched
 241 launch count but was 5–8% slower than the segmented attention microbenchmark, so it was not
 242 selected. Earlier backend-exclusive and split-256 full-decoder pilots did not exceed the 1.1× target.
 243 Fixed-window graph experiments were diagnostic upper bounds because their absolute positions
 244 were not reusable for sustained serving. None of these numbers enters Table 1.

245 Layer diagnostics were used only to understand quality failures. They showed that small local quan-
 246 tization deviations can be amplified by later residual blocks. This observation motivates robustness
 247 allocation but does not prove that the final exact prefix, static suffix, or tail is universally optimal.

248 G Artifact Ledger

	Artifact	SHA-256
	Final quality aggregate	7adbbdc36f14c985004fe22349087210427f3831ee52d92e5a537667c81a64f9
	Final performance analysis	42c3553e3c652601c0fabacb6e59f430fb5589ff5f927c440592d72e584a19fc
249	Explicit-reconstruction diagnostic	20c64fc5b9413a6888287a0c8e6ae29fa2fafbd1c307d6e2d5ef51ea49db367d
	Original source-frozen selection	d1c8675b3b1f70437023dc8cd56665ea79602aaf6eda2752717c0cebc127798f
	Final performance analyzer source	82049fdd00b48cd5f56d9bfd511b3f4b1aa8ddd1fa45f1e20efb7855af60b3a5

250 H Limitations Expanded

251 The algebraic identity applies whenever its center and scalar-scale assumptions hold, but the im-
252 plementation evidence is one Mistral-7B configuration on one RTX 5090/SM120. We have not
253 established transfer to other architectures, models, batch sizes, or context regimes. Code values are
254 converted into FP16 register arithmetic; low-bit tensor-core accumulation is not claimed. Exact re-
255 gions consume 1.508 GiB of FP16 K+V in the final cache, and gross INT8 storage for those pages
256 is retained. The quality policy was repaired after TEST exposure, so a future independently held-
257 out corpus is needed for an unbiased generalization claim. Finally, the timing path is a complete
258 decoder with device-dynamic graph banks and real page planning, but it omits multi-tenant request
259 scheduling, networking, memory admission, and prefill latency.

260 **LLM assistance.** LLM-based coding assistance was used for implementation support, protocol
261 auditing, and manuscript editing. It was not treated as experimental evidence: reported values are
262 bound to deterministic reducers, frozen artifacts, source hashes, and explicit human-reviewable ac-
263 ceptance gates.

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